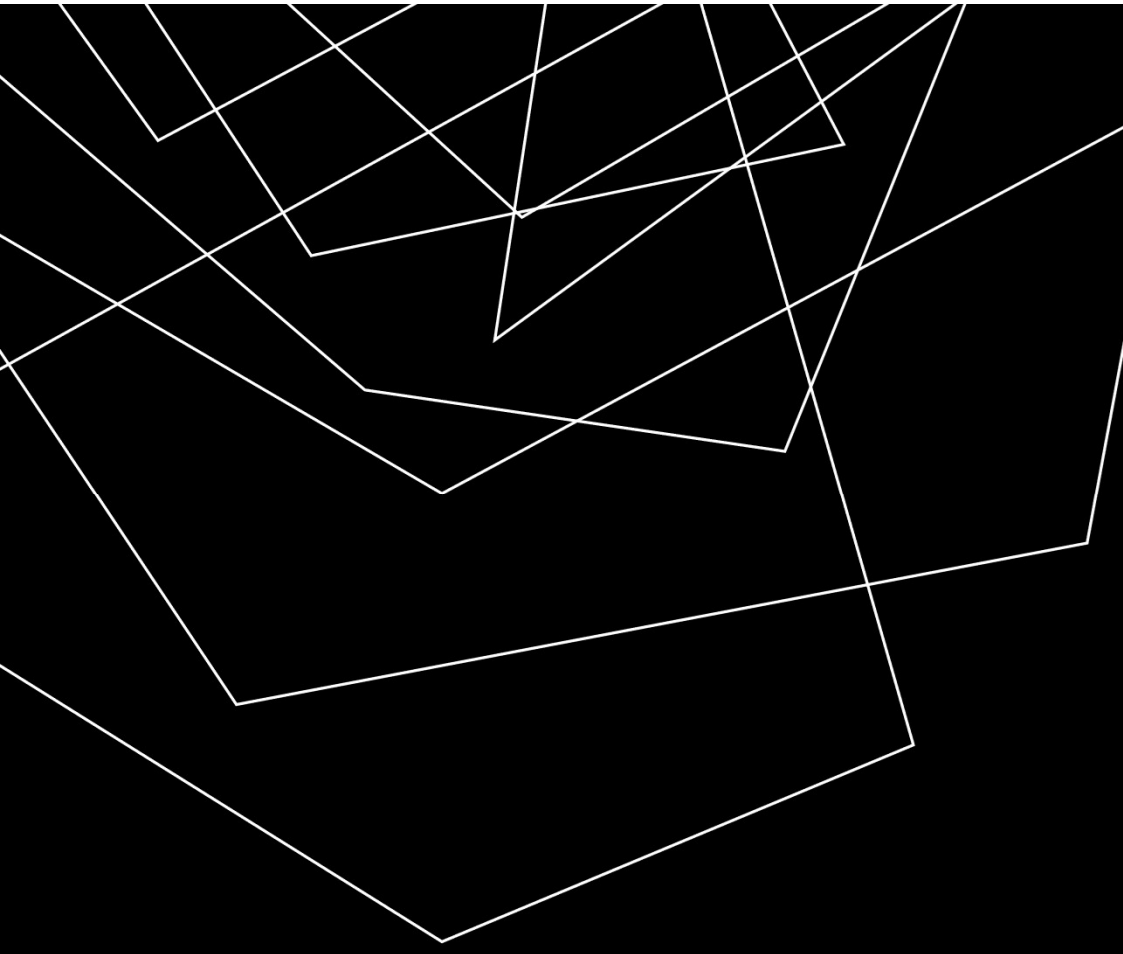




AGENTIC FINANCIAL ADVISOR BOT FOR INVESTMENT STRATEGIES

KEVIN JUANDI



PROJECT TEMPLATE CHOSEN

- 📌 Template: Financial Advisor Bot
- Task: Active Portfolio Management
- Tools: Financial APIs + AI Models + LLM



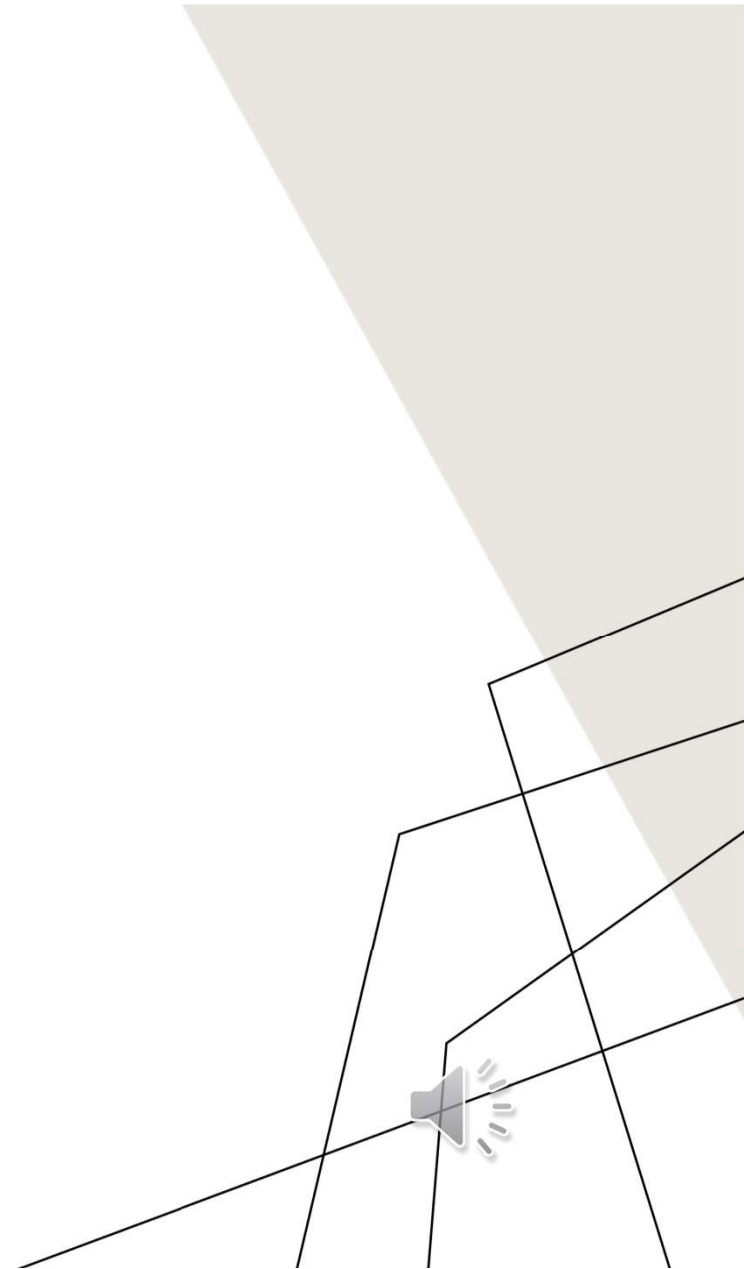
MOTIVATION

Problem:

- ⚠️ Information Overload
- 💰 Costly Advisors
- 🤖 Passive Tools

Users want:

- ✅ Simple advice
- ✅ Transparency
- ✅ Adaptability



RELATED WORKS

Project	Strengths	Weakness
Wealthfront ^[1]	Automated investing	No ML integration
FinRL ^[2]	Open-source, Advanced RL-based trading research	High technical barrier
Finance Wizard ^[3]	NLP integration (ChatGPT)	Blackbox
3Commas ^[4]	Visual bot builder	Rules-based bots

[1] Wealthfront. (n.d.). *Automated investing with Wealthfront*. Retrieved May 8, 2025, from <https://www.wealthfront.com>

[2] Liu, X.Y., Yang, H., Gao, J., & Wang, C. (2021). FinRL: deep reinforcement learning framework to automate trading in quantitative finance. In Proceedings of the Second ACM International Conference on AI in Finance. ACM.

[3] Finance Wizard. (n.d.). *Finance Wizard: Precise Stock Predictions with AI-Driven Insights*. Retrieved May 8, 2025, from <https://financewizard.io/>

[4] 3Commas. (n.d.). *Smart crypto trading terminal*. Retrieved May 8, 2025, from <https://3commas.io>

PROJECT IDEA

Perception

- Pull and preprocess live financial data.
- Yahoo Finance API, Bloomberg API, CCXT

Reasoning

- Apply models to predict outcomes.
- ARIMA, Reservoir Computing

Action Selection

- Choose advisory outputs based on predictions.
- FinRL

Natural Language Reporting

- Pass planning output to LLM to produce clear advice.
- Qwen3

Self-Evaluation

- Log past predictions and compare to real outcomes over time.
- Backtesting

"ALRIGHT, AFTER PULLING THE FRESHEST DATA:

CURRENT AAPL PRICE IS \$209.28 (APRIL 25TH CLOSE).

AFTER-HOURS IT'S SITTING AROUND \$208.60.

52-WEEK RANGE: \$169.11 TO \$260.10.

ANALYST 1-YEAR TARGET ESTIMATE: \$236.23.

🌟 MY PRICE PREDICTION FOR AAPL: WITHIN THE NEXT 3 MONTHS, I PREDICT AAPL WILL RISE TO APPROXIMATELY \$226.45 BARRING MAJOR MARKET DISRUPTIONS.

DISCLAIMER: THIS PREDICTION IS FOR INFORMATIONAL PURPOSES ONLY."



ALGORITHM AND TECH STACK RATIONALE

Predictive Model

- ARIMA (fast and lightweight)
- LSTM vs Reservoir Computing^[5]

FinRL^[2]

- RL environment
- Backtesting

Smolagents^[6]

- Lightweight agentic framework

[5] Kaushal Kumar. (2023). Forecasting Crude Oil Prices Using Reservoir Computing Models.

[2] Liu, X.Y., Yang, H., Gao, J., & Wang, C. (2021). FinRL: deep reinforcement learning framework to automate trading in quantitative finance. In Proceedings of the Second ACM International Conference on AI in Finance. ACM.

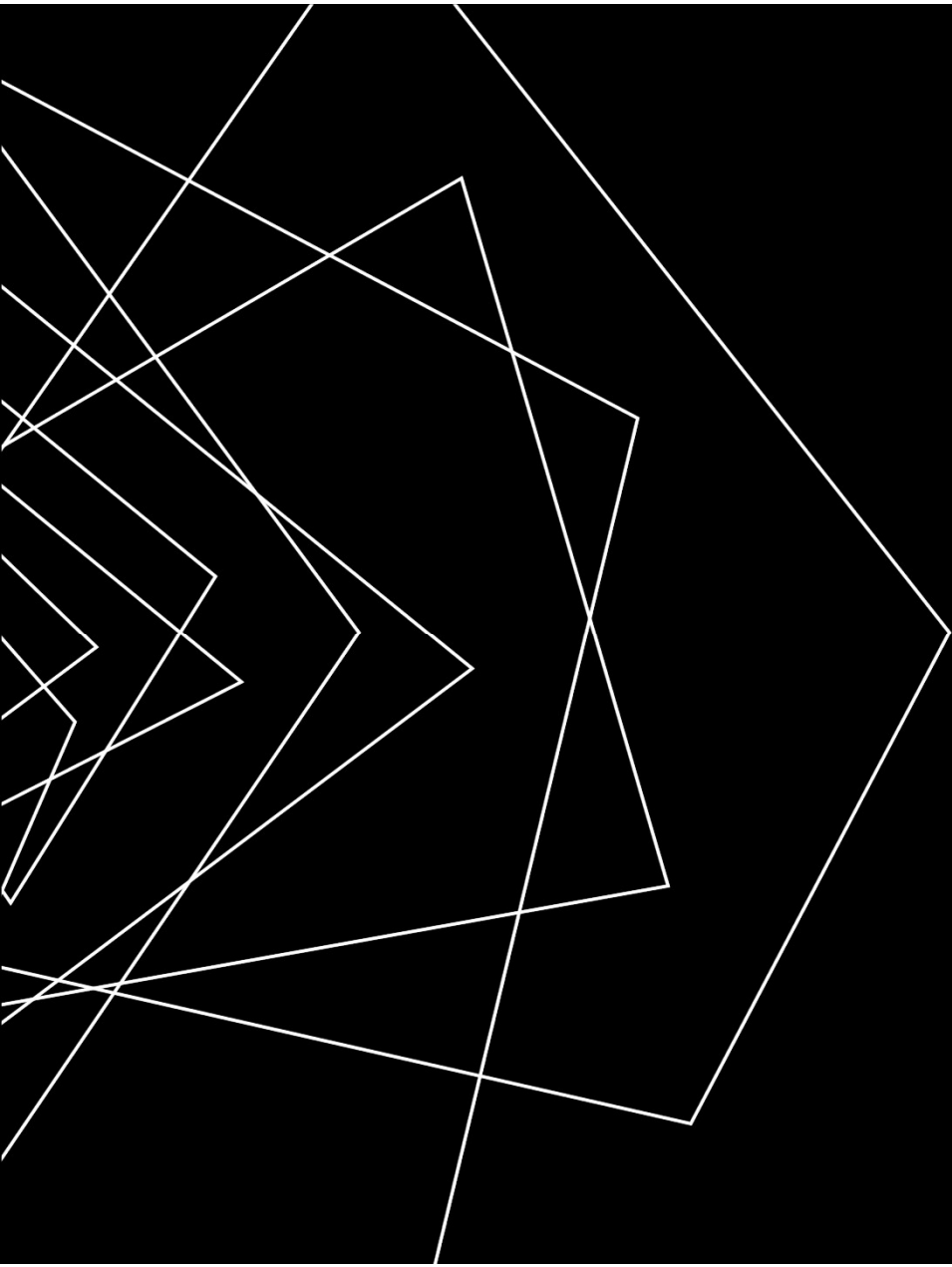
[6] Aymeric Roucher, Albert Villanova del Moral, Thomas Wolf, Leandro von Werra, & Erik Kaunismäki. (2025). `smolagents`: a smol library to build great agentic systems.. .



FINAL IMPACT & VISION

-  Explainable
-  Adaptable
-  Lightweight
-  Expandable to Web or Messaging





THANK YOU

